

Size-specific weighted detectability index for computed tomography image characterization

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Abstract

Background: Technological developments in computed tomography (CT) have increased the diversity of acquisition and reconstruction strategies available for clinical imaging. Task-based image quality metrics that account for patient-size variability are needed to characterize reconstructed image performance and support protocol optimization across clinically relevant imaging conditions.

Purpose: To propose and evaluate a novel size-specific weighted detectability index (SSW-d'), a task-based image performance metric (d') that incorporates lesion contrast weighting and water-equivalent diameter variability for the characterization of CT images acquired at low dose.

Methods: First, a multi-sized image quality phantom was scanned on two CT systems: one photon-counting CT (PCCT, CT1) and one CT equipped with energy integrating detectors (EICT, CT2). Acquisitions were performed with a volume CT dose index (CTDI_{vol}) of 3.2 mGy at 120kV without a tin filter (Sn), at Sn100 and Sn140kV. Second, acquisitions at 3.2 mGy were performed on CT1 using three reconstruction algorithms (FBP, QIR-2, QIR-4). Subsequently, five reconstruction kernels (Br32, Br44, Br56, Br68, and Br76) were evaluated at 1.1 mGy. A higher CTDI_{vol} value of 6 mGy, combined with Br68 and QIR-4, was also tested. The detectability index values (d') were determined for two 10 mm diameter lesions according to two different levels of contrast, based on iodine and bone equivalent tissue. The SSW-d' was calculated for each CT system and for each individual task. Then, SSW-d' was expanded to include both tasks, applying contrast weights determined for each of them.

Results: On CT1, the highest SSW-d' values were 19.3 ± 0.3 and 6.9 ± 0.0 , at 120 kV for bone and iodine rods, respectively. On CT2, the highest SSW-d' results were 19.0 ± 0.3 and 4.9 ± 0.0 at Sn100 kV for bone and iodine rods, respectively. The lowest values were at Sn140kV for both rods and CT systems. When encompassing both tasks, the maximum SSW-d' values were 10.05 ± 0.04 and 8.25 ± 0.03 , respectively obtained at 120 kV for CT1 and at Sn100 kV for CT2. The results showed that the lowest SSW-d' values were at Sn140 kV for both CT systems. Comparison of acquisition and reconstruction parameters showed that the highest SSW-d' value was obtained using the Br32 kernel (17.6 ± 0.7), approximately twofold higher than that obtained at 6 mGy (8.6 ± 0.3). The lowest

Abbreviations: ADMIRE, Advanced Modeled Iterative REconstruction; CT, Computed tomography; CTDI_{vol}, Volume CT dose index; d', Detectability index; D_w, Water equivalent Diameter; EICT, Energy Integrating detectors Computed Tomography; HU, Hounsfield unit; NPS, Noise power spectrum; PCCT, Photon-counting Computed Tomography; QIR, Quantum Iterative Reconstruction; ROI, Region of interest; SD, Standard deviation; SSW-d', Size-Specific Weighted Detectability index; TTF, Task-based Transfer Function; ULD, Ultra-Low dose.

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SSW-d' value (1.9 ± 0.1) was obtained with FBP, while the highest performance was achieved with QIR-4 reconstruction.

Conclusions: A novel metric based on a size-specific weighted detectability index (SSW-d') was developed and evaluated for the characterization of reconstructed CT image performance across a range of patient-equivalent diameters and clinically relevant detection tasks. The proposed SSW-d' consolidates multiple detectability index measurements into a single descriptor that accounts for lesion contrast weighting and patient-size variability. The metric was sensitive to acquisition and reconstruction parameters and proved useful for identifying imaging conditions that maximize task-based performance in ultra-low-dose CT. These findings suggest that SSW-d' may serve as a practical tool for image-quality characterization and protocol optimization across clinically relevant imaging scenarios.

KEYWORDS

computed tomography, detectability index, size-specific weighted detectability, task-based detection, ultra-low dose, water equivalent diameter

1 | INTRODUCTION

For many years, the evaluation of image quality in computed tomography (CT) has prompted research aimed at achieving a balance between image quality and patient exposure to radiation dose for clinical tasks. The detectability index was introduced to provide a standardized method for benchmarking tasks between different CT systems.^{1,2} It quantifies the efficiency with which a specified lesion in the human body can be identified, considering parameters such as image noise, contrast,³ spatial resolution, and object size, given a specific radiation dose.⁴ Then, the balance between radiation dose and image quality can be described using a figure-of-merit (FOM).⁵ Göppel *et al.* proposed a FOM that characterizes image quality in the presence of low-contrast structures in relation to radiation dose using a mathematical observer model.⁶ Different CT scanners and reconstruction algorithms were benchmarked using this model, which makes it useful for type testing CT systems when varying volume CT dose index (CTDI_{vol}). Several studies emphasized the significance of analyzing task-based image quality using multi-sized phantoms, particularly at low doses (LD).^{7,8} However, this calculation method fails to consider variations in slice diameter and phantom attenuation when comparing protocols at fixed radiation dose values. Because fixed-dose protocols remain common for ultra-low-dose (ULD) imaging, comparisons performed at predefined dose levels retain important clinical relevance.

To date, no established protocols at fixed radiation dose values accounts for patient diameter or circumference in specific scanned anatomical regions. CTDI_{vol} selection is currently based solely on patient weight and height. For example, in CT for lung cancer screening, the target CTDI_{vol} is set for a reference patient of standard weight and height, with two additional dose levels for higher and lower BMI values.⁹ Conversely, several

studies reported that mean body diameter and body circumference are more appropriate parameters than patient weight and height for thoracic and pelvic CT studies.^{10,11}

Technological developments in CT have considerably expanded the range of acquisition and reconstruction options available in clinical practice. These advances have increased the need for task-based image quality metrics capable of characterizing image performance under varying imaging conditions and across different patient morphologies and enabling diagnoses with low and ultra-low doses of ionizing radiation.¹² To allow the reduction in patient radiation doses, many methods, such as iterative reconstruction and deep learning-based reconstruction algorithms,¹³ have been implemented.¹⁴ Some CT systems now offer spectral shaping, which involves an additional built-in tin or silver filter that selectively absorbs low-energy photons,¹⁵ and thus hardens the X-ray beam.¹⁶ Furthermore, advanced CT technologies based on semiconductor photon-counting detectors (PCCT) enable the removal of electronic noise, thereby increasing the weight of the low energy photons in the formation of the image. Whereas the high-energy photons contribute more to the final image with the energy integrating CT (EICT), thus affecting the contrast. Several comparative studies of task-generic image quality have been carried out between PCCT and EICT at low doses.^{7,17} Figure-of-merit based studies including a clinically relevant task have shown that, compared with EICT, PCCT reduces image noise and radiation dose, optimizes the contrast-to-noise ratio, and enables access to spectral characteristics.^{8,18–20}

The purpose of this study was first to adapt Göppel's weighted detectability index to include variation in water equivalent diameter at a constant dose level, as in ultra-low dose (ULD) protocols. Based on this approach, a size-specific weighted detectability index

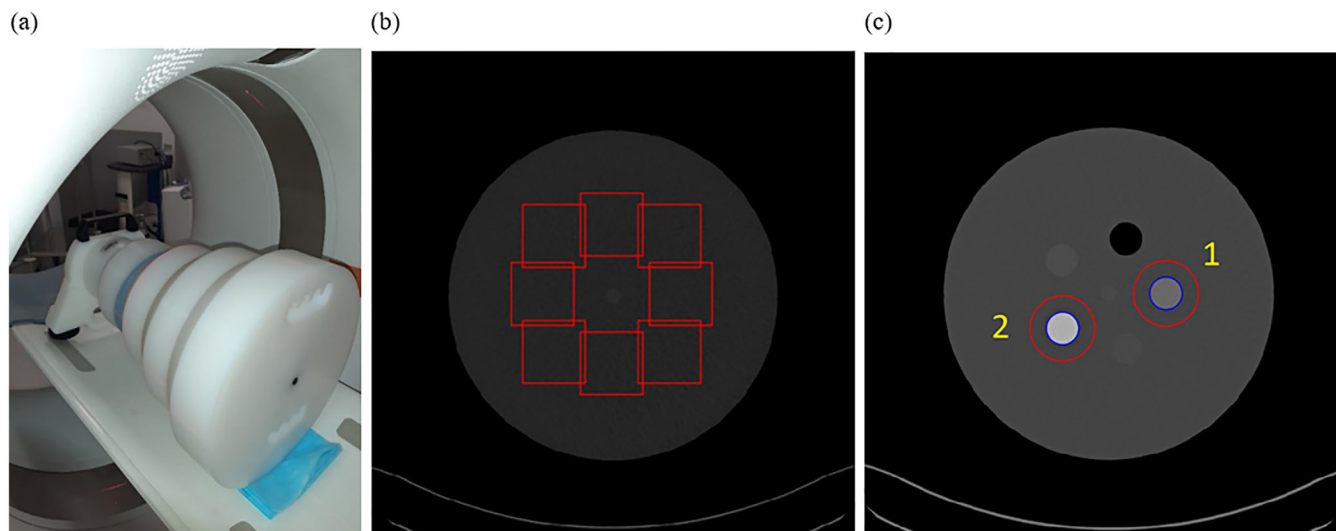


FIGURE 1 (a) Photograph of the Mercury v4.0 multi-sized phantom showing the diameter variation according to five sections. Only the three largest ones were part of the study. (b) CT image of the 26 cm section showing the eight regions of interest used to evaluate the noise power spectrum, highlighted in red. (c) CT image of the 26 cm section used to calculate the task-based transfer function with the iodine (1) and bone (2) rods.

(SSW-d') was developed as a metric for CT image characterization across a range of patient-equivalent sizes. Second, the proposed metric was evaluated using two CT systems employing different detector technologies (PCCT and EICT) in order to investigate its ability to characterize reconstructed image performance for clinically relevant detection tasks. Finally, the influence of dose level, reconstruction algorithm, and reconstruction kernel on SSW-d' was investigated for ultra-low-dose pelvis CT protocol.

2 | MATERIALS AND METHODS

2.1 | Phantom and determination of the water equivalent diameters

Acquisitions were made on a multi-sizes phantom (Mercury v4.0, Sun Nuclear, Melbourne, FL, USA). The three largest sections of the phantom (26, 31, 36 cm in diameter) were used to perform the task-based image quality assessment (Figure 1a). Each section was composed of a homogeneous ultra-high molecular weight polyethylene part and a part that contains five rods: solid water, air, 10 mg/mL iodine, polystyrene, and bone. The homogeneous part was used to compute the noise power spectrum (NPS; Figure 1b) and the water equivalent diameters. The part containing the five rods was used for the task-based transfer function calculation (TTF). In this study, only the iodine and bone rods were used. They were chosen to represent different levels of contrast to evaluate the performance of a CT system. (Figure 1c).

The nominal phantom diameters were 26 cm, 31 cm, and 36 cm, respectively. As the documentation

did not specify the uncertainty regarding the phantom diameters, measurements were taken from the images acquired for each one. Ten measurements were carried out on a single slice of the homogeneous section for each of the phantom diameters. The water equivalent diameters (D_w) were calculated according to AAPM Report No. 220 and determined for each section as follows:²¹

$$D_w = 2 \sqrt{\left[\frac{1}{1000} \overline{CT(x, y)_{ROI}} + 1 \right] \frac{A_{ROI}}{\pi}} \quad (1)$$

where $\overline{CT(x, y)_{ROI}}$ was the mean CT number of pixels in the circular ROI placed in the whole homogenous section of the phantom made in polyethylene, A_{ROI} was the area of each phantom section of one slice.

D_w values were calculated for each CT system and for each phantom section, corresponding to water-equivalent diameters D1, D2, and D3.

2.2 | CT scanners and scanning protocols

2.2.1 | Low-dose image quality characterization across two CT systems

Acquisitions were performed on two Siemens Healthineers CT systems, which represent different technical specifications: the NAEOTOM Alpha.Peak (CT1; European Georges Pompidou Hospital), the SOMATOM Edge (CT2; Nîmes University Hospital). Photon-counting detectors were used in CT1, while CT2 was equipped

TABLE 1 Acquisition and reconstruction parameters used on the two CT systems.

CT system	CT1	CT2
Beam collimation (mm)	144 × 0.4	128 × 0.6
Tin filter thickness (mm)	0.4	0.6
Detector type	Photon counting	Energy integrating
Data acquisition		
Tube voltage (kV)/mAs/Displayed CTDI _{vol} (mGy)	120/ 40/ 3.16	120/ 47/ 3.20
	Sn100/ 375/ 3.20	Sn100/ 791/ 3.20
	Sn140/ 90/ 3.21	Sn140/ 143/ 3.20
Pitch factor at 120/Sn100/Sn140	1/ 1/ 1	0.95/ 0.4/ 1
Rotation time (s)	0.5	0.5
Field of View (FOV)	380 mm	
Matrix size	512 × 512	
Image reconstruction		
Reconstruction Software Version	Syngo CT VB10A	Syngo CT VB20A
Convolution Kernel	Br68	Br62
Reconstruction algorithm	QIR-4	ADMIRE-5

with energy integrating detectors. Each of them was equipped with a tin filter (Sn) with different thicknesses: 0.4 mm for CT1 and 0.6 mm for CT2 (Table 1).

Acquisitions were performed for all CT systems at Sn100 kV, Sn140 kV, and 120 kV. These three tube voltages were selected as they were specific for optimized protocols. For ultra-low-dose protocols, Sn100 kV and Sn140 kV are most commonly used for standard-sized and obese patients, respectively.²² A comparison was made between the image quality achieved using a tin filter and a tube voltage of 120 kV, which was used as a reference setting.

For each CT acquisition, the tube current modulation system was disabled. The tube current values were adapted to achieve a CTDI_{vol} close to 3.2 mGy, which is considered in clinical routine as a low-dose level for lumbar spine and pelvis CT examination.²³

The other parameters of the acquisition are described in Table 1. For each tube voltage and each CT system, ten repeated acquisitions were performed for each setting.

A bone reconstruction kernel used for ultra-low-dose lumbar spine CT protocols was studied for each CT system. Each one was specific to each CT system that was studied. The reconstruction kernels were Br68 and Br62 for CT1 and CT2, respectively. Raw data for CT1 were reconstructed using Quantum Iterative Reconstruction (QIR) level 4 and Advanced Modeled Iterative Reconstruction (ADMIRE) level 5 for CT2. These were the highest levels available for each algorithm. All images

were reconstructed as polychromatic images (T3D) using data from all detected events above the lowest energy threshold at 20 keV.²⁴

All CT images used a 380 mm field of view, 512 × 512 pixels matrix size, and 1 mm slice thickness with 1 mm slice increment.

2.2.2 | Ultra-low-dose protocol definition

Further characterization of the CT1 image quality was performed across varying acquisition and reconstruction parameters to determine the optimal parameters for ULD image quality. Each parameter was then varied individually, while all other parameters were kept constant, to isolate its specific impact on image quality.

A reference protocol, defined in the first part of the study as a low-dose (LD) protocol (Sn 140 kV, 90 mAs, Br68 kernel, QIR-4, CTDI_{vol} = 3.2 mGy), was used as the baseline. This protocol was compared with two alternative reconstruction algorithms, Filtered Back Projection (FBP) and QIR-2, with CTDI_{vol} held constant. In addition, a standard-dose acquisition (Sn 140 kV, 169 mAs, Br68 kernel, QIR-4, CTDI_{vol} = 6.0 mGy) was included for comparison. An ultra-low-dose protocol (Sn 140 kV, 31 mAs, QIR-4, CTDI_{vol} = 1.1 mGy) was also evaluated, based on the methodology described by Stern *et al.*²⁵ The impact of five reconstruction kernels: Br32, Br44, Br56, Br68, Br76, was assessed for this ULD protocol at 1.1 mGy, all corresponding to standard body kernels. All images were reconstructed as polychromatic images with a 380 mm field of view, 512 × 512 matrix, 1-mm slice thickness, and 1-mm slice increment.

2.3 | Task-based image quality assessment

A task-based image quality assessment was performed using the iQMetrix-CT software (software version: v1.2).²⁶ NPS, TTF and detectability indexes (*d'*) were computed for two groups of five consecutive acquisitions performed on each CT system, tube voltage, and diameter.

2.3.1 | Noise power spectrum

The NPS was computed from two groups of five repeated acquisitions on 20 consecutive axial slices by placing eight square regions of interest (ROIs) in a homogeneous part of each diameter section (Figure 1b) as follows (1):

$$\begin{aligned}
 NPS_{2D}(f_x, f_y) &= \frac{\Delta_x \Delta_y}{L_x L_y} \frac{1}{N_{ROI}} \sum_{i=1}^{N_{ROI}} |FFT_{2D}\{ROI_i(x, y) - FIT_i(x, y)\}|^2 \\
 &\quad (2)
 \end{aligned}$$

where Δ_x and Δ_y are the pixel size in the x- and y-directions, respectively; *FFT* is the Fast Fourier Transform; L_x and L_y are the lengths of the ROIs in the x- and y-directions; N_{ROI} is the number of ROIs; $ROI_i(x, y)$ is the mean pixel value measured of a ROI at the position (x, y) and $FIT_i(x, y)$ is a 2nd order polynomial fit of $ROI_i(x, y)$.

The ROI sizes were adjusted to D1, D2 and D3: 66 × 66 pixels, 90 × 90 pixels and 114 × 114 pixels, respectively.

To quantify the changes in magnitude and texture of the noise, the square root of the area under the NPS2D curve and the weighted average spatial frequency (f_{av} , mm⁻¹) were measured.⁵

2.3.2 | Task-based transfer function

The Task-based transfer function (TTF) was calculated on the iodine and bone rods (Figure 1c) using the circular edge technique.^{27,28} To minimize the image-noise effect on the edge spread function, the TTF was computed from two groups of five repeated acquisitions on 100 consecutive axial slices (20 slices for each of the 5 groups of acquisitions).²⁹ The contrast between the rods and the phantom background material was computed during the TTF calculation process and studied.

2.3.3 | Detectability index

A non-prewhitening observer model with an eye filter was used to calculate the detectability index d' :

$$d'^2 = \frac{\left[\iint |W(u, v)|^2 \cdot TTF(u, v)^2 \cdot E(u, v)^2 du dv \right]^2}{\iint |W(u, v)|^2 \cdot TTF(u, v)^2 \cdot NPS(u, v) \cdot E(u, v)^4 du dv} \quad (3)$$

where u and v are spatial frequencies in the x- and y-directions, E is the eye filter that models the human visual system sensitivity to different spatial frequencies,^{30–34} and $W(u, v)$ the task function defined as:

$$W = |F\{h_1(x, y) - h_0(x, y)\}| \quad (4)$$

where F the Fourier transform, $h_1(x, y)$ and $h_0(x, y)$ correspond to the object present and the object absent hypotheses, respectively.⁵

The eye filter was modeled according to the visual response function using a display size of 153.6 mm, a 400-mm viewing distance,³⁵ and the Saunders visual function.³⁶ Two task functions of 10-mm diameter were considered, and the task contrast was defined as the rod-background CT number difference measured during the TTF calculation process for each configuration.

To characterize the variability of the detectability index (d') across phantom diameters and assess its

dependence on patient diameter, the mean, minimum, maximum, and inter-diameter coefficient of variation (CV) of d' were calculated. The inter-diameter CV was defined as the ratio of the standard deviation to the mean and was used to quantify the relative variability of d' with patient-equivalent diameters.

2.4 | Calculation of the Size-Specific Weighted Detectability index of reconstructed CT images

The process of determining Size-Specific Weighted Detectability index (SSW- d') was as follows: two contrast rods (iodine and bone rods) with different contrast levels (C_{Bone} and C_{Iodine}) were imaged at a single CTDI_{vol} dose level and for three phantom water equivalent diameters (D1, D2 and D3).

For each phantom water equivalent diameter D_i ($i = 1$ to 3), detectability indices were calculated for both rods, yielding $d'_{i,Iodine}$ and $d'_{i,Bone}$. These values were then averaged across rod types and phantom sizes according to an approach adapted from the method of Goppel et al⁶:

$$\langle SSW - d' \rangle_{Iodine} = \frac{1}{2(D_3 - D_1)} \sum_{i=1}^3 K_i \cdot d'_{i,Iodine} \quad (5)$$

$$\langle SSW - d' \rangle_{Bone} = \frac{1}{2(D_3 - D_1)} \sum_{i=1}^3 K_i \cdot d'_{i,Bone} \quad (6)$$

where $K_1 = D_2 - D_1$, $K_2 = D_3 - D_1$ and $K_3 = D_3 - D_2$ are the diameter weights for the corresponding values of d'_{Iodine} and d'_{Bone} .

The mean SSW- d' across the two contrast rods was calculated as follows:

$$SSW - d' = \Omega_{Iodine} \cdot \langle SSW - d' \rangle_{Iodine} + \Omega_{Bone} \cdot \langle SSW - d' \rangle_{Bone} \quad (7)$$

The weighting factors Ω_{Iodine} and Ω_{Bone} were defined based on the inverse rod contrasts as follows:

$$\Omega_{Bone} = \frac{C_{Bone}^{-1}}{C_{Bone}^{-1} + C_{Iodine}^{-1}} \quad (8)$$

$$\Omega_{Iodine} = \frac{C_{Iode}^{-1}}{C_{Bone}^{-1} + C_{Iodine}^{-1}} \quad (9)$$

where C_{Bone} and C_{Iodine} are the contrast in Hounsfield Units of the bone and iodine rods, respectively.

This formulation was based on the approximately linear relationship between detectability and rod contrast at a fixed rod diameter. Equations (8) and (9) provide a reliable estimate of the weights, helping to strike a balance between the various inserts

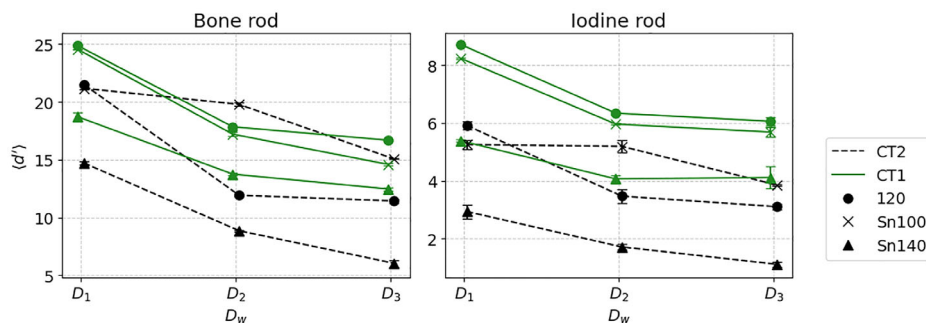


FIGURE 2 Detectability index d' according to the phantom water equivalent diameter (D_w), bone and iodine rods, for several tube voltages on the two CT systems. The task function of 10-mm diameter detection was considered.

and preventing the easily detectable ones from dominating.

2.5 | Uncertainty of the Size-Specific Weighted Detectability index

The uncertainty of mean $SSW-d'$ was estimated by propagating the uncertainties of its components, including $\langle SSW - d' \rangle_{Bone}$, $\langle SSW - d' \rangle_{Iodine}$, and the corresponding weighting factors Ω_{Iodine} and Ω_{Bone} . It was assumed that the uncertainties of d' values were identical across water-equivalent phantom diameters, and that the uncertainties of K values were constant.

Finally, the uncertainty of $SSW-d'$ including the two rod contrasts was defined as follows:

$$u(SSW - d')^2 \sim \left(\Omega_{Bone} \times u(\langle SSW - d' \rangle_{Bone}) \right)^2 + \left(\Omega_{Iodine} \times u(\langle SSW - d' \rangle_{Iodine}) \right)^2 \quad (10)$$

2.6 | Data analysis

All results were reported as the mean $\pm$ expanded uncertainty, with the expanded uncertainty calculated using a coverage factor $k = 2$. This corresponded to a confidence level of approximately 95%. Descriptive statistics are also reported as median percentages with their interquartile ranges [IQR, 25th–75th percentiles]. Group differences were assessed using the Wilcoxon signed-rank test for paired data, with a significance level of 0.05.

3 | RESULTS

3.1 | CT image quality at low dose across two CT systems

The results of the noise magnitude, average NPS spatial frequency, contrast values and TTF at 50% are

shown in Table S1 and Table S2, respectively. The mean water equivalent diameters for the two CT systems were 246.5 ± 0.42 mm, 294.3 ± 0.14 mm and 341.8 ± 0.57 mm for D1, D2 and D3, respectively.

3.1.1 | Influence of water equivalent diameters on the detectability index between CT systems

Figure 2 shows the detectability index values for the two CT systems using bone and iodine rods for three tube voltages according to the phantom's water-equivalent diameter.

For both CT systems, there was a decrease in detectability index with an increase in water equivalent diameter from D1 to D3 at a fixed low-dose level ($p = 0.031$). However, the decrease in detectability index with increasing water-equivalent diameter was more pronounced on CT2 than on CT1, as reflected by higher inter-diameter CVs across phantom diameters for both the bone rod ($32.4 \pm 2.6\%$ vs $21.2 \pm 0.2\%$) and the iodine rod ($39.3 \pm 3.7\%$ vs $24.3 \pm 0.8\%$), indicating a stronger dependence of image performance on patient-equivalent diameters (Table 2). This finding highlights the substantial influence of phantom size on task-based image performance. For each water equivalent diameter, the minimum detectability index values were at Sn140 kV.

The detectability index values on CT1 were higher than those on CT2. When including all tube voltages and diameters, the median detectability index increases on CT1 compared to CT2 were: 44.0 % [IQR, 32.8 %–49.2 %] and 21.4 % [IQR, 13.4 %–33.2 %] for iodine and bone rods, respectively. This difference was statistically significant ($p < 0.05$).

For the CT1 system, the highest detectability index values were obtained at 120 kV whatever the phantom water equivalent diameter and both rods. For the CT2 system, the highest detectability index values were obtained at 120 kV for D1 and Sn100 kV for D2 and D3, for the two rods. Notably, similar results were observed at 120 and 100 kV for the iodine rod on CT1 for all diameters, as well as for the bone rod

TABLE 2 SSW-d' performance across radiation dose levels, reconstruction algorithms, and reconstruction kernels. The coefficient of variation across diameters (inter-diameter CV) of the detectability index (d') is reported for bone and iodine rods, together with the corresponding d' across varying phantom diameters. Size-specific weighted detectability (SSW - d') is also reported for individual rods and for all rods, with results expressed as mean $\pm$ expanded uncertainty.

Parameters		Bone rod			Iodine rod			SSW-d'
		Mean d' (min-max)	Inter-diameter CV (%)	$\langle \text{SSW} - d' \rangle_{\text{Bone}}$	Mean d' (min-max)	Inter-diameter CV (%)	$\langle \text{SSW} - d' \rangle_{\text{Iodine}}$	
Displayed CTDI_{vol} <i>Br68, QIR-4</i>	1.1 mGy	12.7 $\pm$ 0.5 (12.2–13.2)	4	12.7 $\pm$ 0.1	3.6 $\pm$ 0.2 (3.4–3.8)	6	3.5 $\pm$ 0.0	5.6 $\pm$ 0.2
	3.2 mGy	17.3 $\pm$ 5.3 (13.0–23.2)	31	16.9 $\pm$ 0.2	4.8 $\pm$ 1.5 (3.6–6.5)	30	4.7 $\pm$ 0.0	7.5 $\pm$ 0.3
	6.0 mGy	19.6 $\pm$ 4.8 (14.6–24.3)	25	19.7 $\pm$ 0.3	5.4 $\pm$ 1.6 (3.9–7.0)	29	5.3 $\pm$ 0.0	8.6 $\pm$ 0.3
Reconstruction Algorithm <i>3.2 mGy, Br68</i>	FBP	4.3 $\pm$ 0.6 (3.6–4.8)	14	4.3 $\pm$ 0.02	1.2 $\pm$ 0.2 (1.0–1.3)	16	1.2 $\pm$ 0.0	1.9 $\pm$ 0.1
	QIR-2	7.4 $\pm$ 1.1 (6.2–8.5)	16	7.4 $\pm$ 0.04	2.2 $\pm$ 0.4 (1.7–2.4)	16	2.2 $\pm$ 0.0	3.4 $\pm$ 0.1
Reconstruction Kernel <i>1.1 mGy, QIR-4</i>	Br32	40.3 $\pm$ 9.3 (32.2–50.5)	23	39.7 $\pm$ 1.3	11.2 $\pm$ 2.8 (8.6–14.2)	26	11.0 $\pm$ 0.1	17.6 $\pm$ 0.7
	Br44	20.4 $\pm$ 4.5 (16.6–25.3)	22	20.1 $\pm$ 0.3	5.3 $\pm$ 1.3 (4.1–6.6)	24	5.2 $\pm$ 0.0	8.6 $\pm$ 0.3
	Br56	16.6 $\pm$ 2.7 (14.1–19.4)	16	16.5 $\pm$ 0.2	4.4 $\pm$ 0.8 (3.7–5.3)	19	4.4 $\pm$ 0.0	7.2 $\pm$ 0.3
	Br76	35.7 $\pm$ 2.3 (33.4–38.0)	6	36.3 $\pm$ 1.0	7.5 $\pm$ 0.6 (7.0–8.2)	9	7.4 $\pm$ 0.0	14.0 $\pm$ 0.6

Note: The reference protocol (bold) was defined as QIR4 reconstruction algorithm with Br68 kernel at CTDI_{vol} = 3.2 mGy.

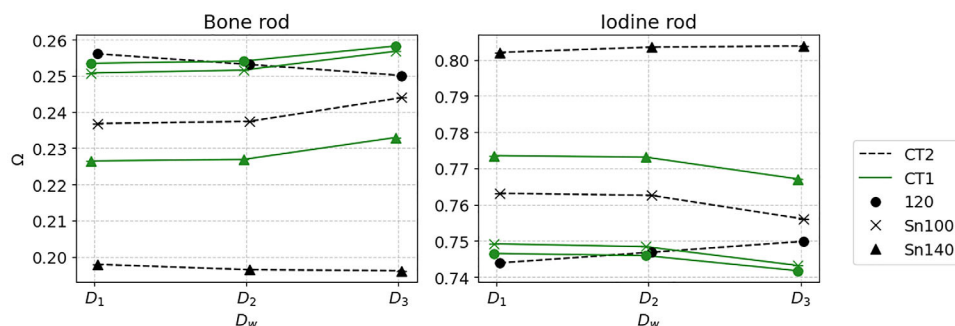


FIGURE 3 Weights (Ω_j) according to water equivalent diameters for bone and iodine rods. Three tube voltages and two CT systems were considered.

with the smallest water equivalent diameter in both systems.

3.1.2 | Influence of water equivalent diameters on contrast weights Ω_j between CT systems

The contrast weight values of the detectability index for the two CT systems are shown in Figure 3, considering bone and iodine rods for three tube voltages according to phantom water equivalent diameters.

The phantom water equivalent diameter increase had no or very slight impact on contrast weights for

both CT system and all the rods. On CT1, identical contrast weights were found at 120 kV and Sn100 kV for each rod. Then, the results showed a very slight decrease in Ω_{Bone} , as well as an increase in Ω_{Iodine} at Sn140 kV ($p < 0.05$). On CT2, the Ω_{Bone} values decreased from 120 kV to Sn140 kV, while the Ω_{Iodine} values increased. The lowest Ω_{Bone} values and the highest Ω_{Iodine} values were reached at Sn140 kV for all phantom water equivalent diameters on CT2.

The results showed that the contrast weight values were lower for the bone rod than for the iodine rod. Specifically, the Ω_{Bone} values were 0.24 ± 0.04 and the Ω_{Iodine} values were 0.76 ± 0.04 ($p < 0.05$) for all

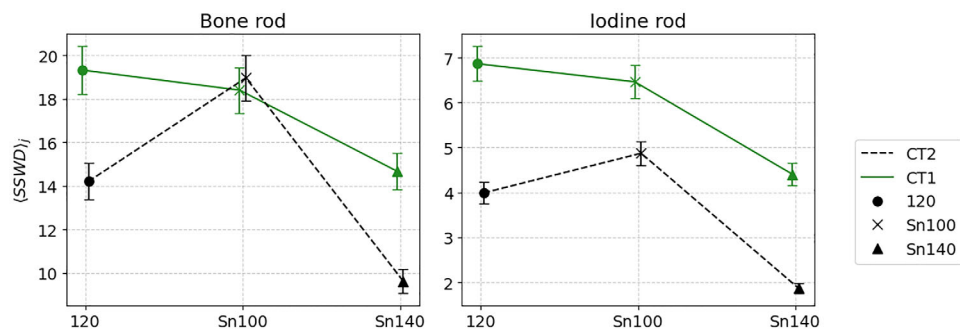


FIGURE 4 Size-Specific Weighted Detectability index $\langle SSW - d' \rangle_i$ for bone and iodine rods, separately, for several tube voltages on the two CT systems. One task function of 10-mm diameter detection was considered.

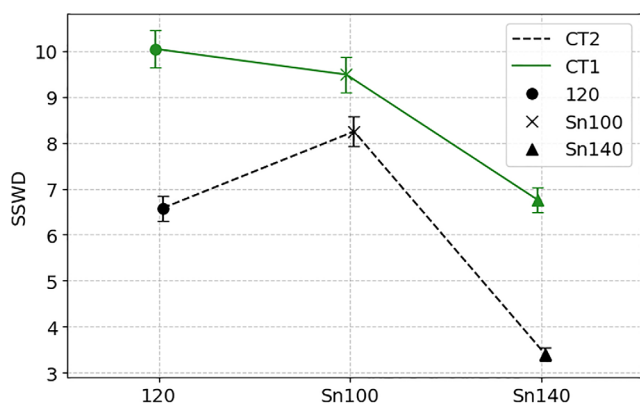


FIGURE 5 Size-Specific Weighted Detectability index ($SSW - d'$) for several tube voltages on the two CT systems when considering all rods. One task function of 10-mm diameter detection was considered. Three water equivalent diameters and two rods were included.

tube voltages and diameters and significantly different between the two CT systems ($p < 0.05$).

3.1.3 | Size-Specific weighted detectability index between CT systems

The water equivalent diameter-weighted detectability index values for the two CT systems are shown on Figure 4 and Figure 5 at several tube voltages.

- Size-Specific Weighted Detectability index according to individual rods

In Figure 4, Size-Specific Weighted Detectability index $\langle SSW - d' \rangle$ results are shown according to each rod.

The results showed that $\langle SSW - d' \rangle_{\text{Iodine}}$ values were about three times lower than $\langle SSW - d' \rangle_{\text{Bone}}$ values ($p < 0.05$).

On CT1, $\langle SSW - d' \rangle$ values decreased from 120 kV to Sn140 kV for both rods. The highest $\langle SSW - d' \rangle$ values were 19.3 ± 0.3 at 120 kV for bone rod and 6.9 ± 0.0 at 120 kV for iodine rod. On CT2, the highest $\langle SSW - d' \rangle$ results for the two rods were 19.0 ± 0.3 and 4.9 ± 0.0

at Sn100 kV for $\langle SSW - d' \rangle_{\text{Bone}}$ and $\langle SSW - d' \rangle_{\text{Iodine}}$, respectively. The lowest values were at Sn140 kV for both rods and CT systems.

The $\langle SSW - d' \rangle_{\text{Bone}}$ values showed good consistency for the two CT systems at Sn100 kV: 18.4 ± 0.3 and 19.0 ± 0.3 for CT1 and CT2, respectively, and were consistent within experimental uncertainty.

- Size-Specific Weighted Detectability index including all rods

In Figure 5, when all inserts were considered, the $SSW - d'$ results differed between the two CT systems, regardless water equivalent diameter.

The maximum $SSW - d'$ values were 10.05 ± 0.04 at 120 kV for CT1 and 8.25 ± 0.03 at Sn100 kV for CT2. The results showed that the lowest $SSW - d'$ values were at Sn140 kV for both CT systems.

4 | CT image quality at ultra-low dose

The results of the noise magnitude, average NPS spatial frequency, contrast values and TTF at 50% are shown in Table S3 and Table S4, respectively.

Table 2 summarizes the results for different parameter configurations with respect to the size-specific weighted detectability $SSW - d'$ for defining an ULD protocol. Figure 6 illustrates the two parameter combinations that yielded the highest $SSW - d'$ values at ULD with the highest and lowest coefficient of variation.

Table 2 shows that compared with the reference protocol (3.2 mGy, Br68, QIR-4), the closest $SSW - d'$ at ULD was observed for 1.1 mGy using the Br56 kernel and QIR-4 reconstruction. The size-specific weighted detectability in the ULD protocol was 4% lower than that of the reference protocol. The differences for individual rods were -2.4 % and -6.4 %, for $\langle SSW - d' \rangle_{\text{Bone}}$ and $\langle SSW - d' \rangle_{\text{Iodine}}$, respectively. The coefficient of variation of d' was higher for the reference protocol than for the ULD protocol, with increases of +48.4% and +36.7% for bone and iodine rods, respectively.

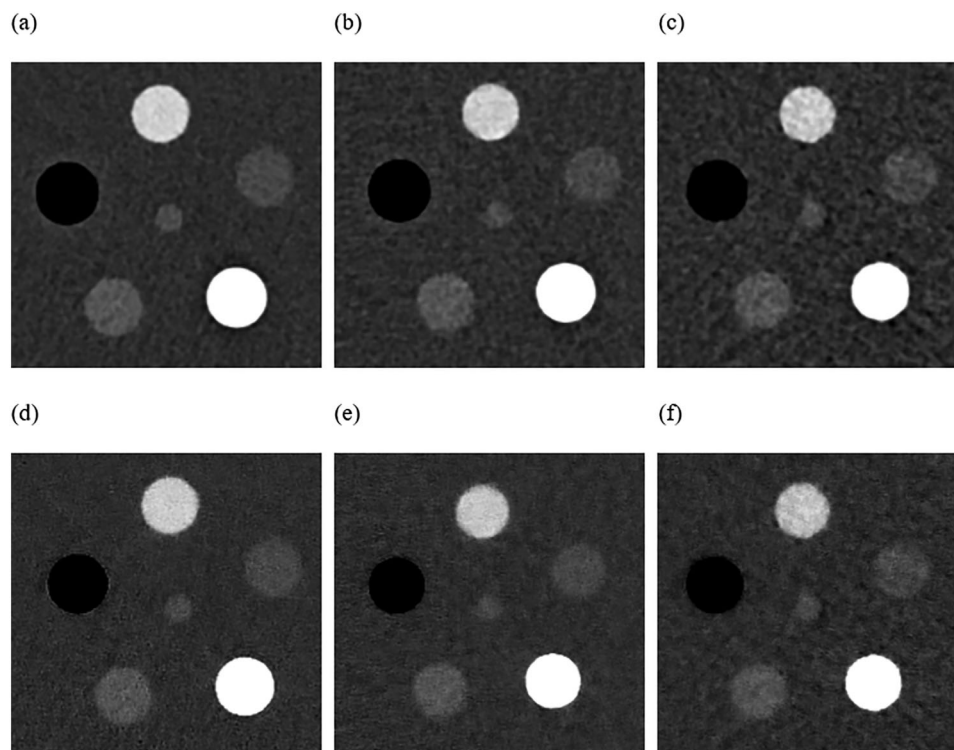


FIGURE 6 Illustration of image quality variability across phantom diameters for the parameter combinations yielding the highest SSW- d' values at ULD. Axial CT images reconstructed for 26 cm (a, d), 31 cm (b, e) and 36 cm (c, f) diameter phantoms using two kernels: Br32 (top row, highest coefficient of variation), and Br76 (bottom row, lowest coefficient of variation). Acquisition parameters were Sn140 kV, $CTDI_{vol} = 1.1$ mGy with QIR4 reconstruction algorithm.

The lowest SSW- d' value 1.9 ± 0.1 was for the FBP reconstruction algorithm, while the highest performance was achieved with QIR-4 in the reference protocol.

The highest SSW- d' value was obtained at ULD using the Br32 kernel (17.6 ± 0.7), approximately twofold higher than that obtained at 6 mGy: 8.6 ± 0.3 . Increasing the radiation dose from 3.2 to 6 mGy had only a limited effect on the SSW- d' value compared with the influence of the reconstruction kernel at lower dose levels.

The maximum d' values for bone and iodine rods were 50.5 and 14.2, respectively, at Br32. However, this configuration showed a high variability across phantom diameters, with an inter-diameter coefficient of variation ranging from 23% to 26% (a -c, Figure 6). In contrast, Br76 yielded a lower maximum d' (38.0 and 8.2 for bone and iodine rods, respectively), but with improved stability, as indicated by a lower coefficient of variation (6%–9%), resulting in an SSW- d' value of 14.0 ± 0.6 (e-f, Figure 6).

At 1.1 mGy, lower inter-diameter coefficients of variation were observed for the Br68 and Br76 reconstruction kernels, indicating more uniform detectability values across the investigated phantom diameters and thus reduced diameter-dependent variability under these acquisition and reconstruction settings.

5 | DISCUSSION

In the present study, SSW- d' was introduced as a task-based metric designed to incorporate phantom diameter into the assessment of CT image quality. To our knowledge, this is the first detectability index-based metric to incorporate phantom size, as represented by water-equivalent diameter, into a single task-based assessment approach. Unlike conventional detectability index analyses, which are performed for specific phantom sizes, SSW- d' aggregates measurements across a range of patient-equivalent diameters and lesion contrast levels. Consequently, SSW- d' offers a more comprehensive description of imaging performance at ultra-low dose, reflecting the combined impact of phantom diameter, acquisition parameters, reconstruction algorithms and kernels, detector technology and task characteristics. The efficiency of the methodology stems from improved interpretability and comparability rather than from a reduction in experimental workload. This approach may be useful for both individual and multi-task analyses and facilitates optimization of low-dose and ultra-low-dose protocols at fixed radiation dose.

This model was a variation of the FOM suggested by Göppel et al.,⁶ characterized as a dose-efficiency index with low contrast structures. While the

detectability index is influenced by acquisition and reconstruction parameters, it is also strongly affected in clinical practice by patient-equivalent diameters and tissue attenuation, through their impact on noise and automatic tube current modulation.³⁷ This highlights the importance of incorporating patient-dependent effects, including variations in patient-equivalent diameter, into objective functions for CT protocol optimization.

Comparative studies in the literature frequently report detectability indices separately for each phantom diameter. However, extrapolating these results to other diameters without accounting for the continuous variation in scatter, attenuation, and noise may lead to oversimplified or potentially misleading conclusions about reconstructed image performance.^{8,38,39} This underscores the importance of the metric that considers phantom diameter when evaluating detectability across imaging conditions.

A similar limitation has long been recognized in CT dosimetry, where CTDI_{vol} alone does not account for patient size and attenuation characteristics. To overcome this issue, the American Association of Physicists in Medicine (AAPM) introduced size-specific conversion factors based on water-equivalent diameter, leading to Size-Specific Dose Estimate (SSDE).⁴⁰ Unlike the physical diameter, the water-equivalent diameter incorporates both patient size and X-ray attenuation properties, providing a more representative description of imaging conditions.

The study results showed that detectability index exhibits a similar dependence on phantom diameter: across all imaging conditions, d' consistently decreased with increasing water-equivalent diameter, reflecting the combined effects of greater attenuation and scatter, which increase image noise and reduce object contrast. The inter-diameter coefficients of variation reported in this study quantified the magnitude of these size-dependent variations in the detectability index. Consistent with these findings, the water-equivalent diameter had a strong influence on d' across all imaging conditions. Although the relative contributions varied across reconstruction settings, the resulting variations were similar in magnitude to those associated with changes in reconstruction parameters in several cases. This emphasizes the importance of phantom diameter as a major determinant of reconstructed image performance under low- and ultra-low-dose conditions. This behavior was consistently observed across the two CT systems and imaging protocols investigated.

The proposed SSW- d' metric addresses this dependence by using water-equivalent diameters combined with weights to adjust the detectability index. The weighting coefficients K_i are explicitly defined by the differences between adjacent water-equivalent diameters, and contribute proportionally to the interval they represent within the range of phantom diameters. At the extremities of the range, the weights are defined using

one-sided differences, while interior weights are based on symmetric differences between adjacent diameters. This ensures a consistent treatment of boundary points and reflects the higher effective sampling density of intermediate diameters in the discrete integration scheme.

The reconstruction kernel analysis at ULD illustrated this effect. Although the Br32 kernel had the highest detectability index, it was associated with reduced robustness across phantom diameters, as indicated by the high inter-diameter coefficients of variation. In contrast, kernels such as Br76 provided more stable performance across phantom sizes, albeit with lower peak detectability. SSW- d' reduced the influence of exceptionally high d' values observed at specific diameters by incorporating weights on detectability over the full range of investigated diameters, providing a more balanced assessment of reconstructed image performance. These findings suggest that protocol optimization strategies for ultra-low-dose CT based solely on maximum detectability may be suboptimal in clinical practice, where consistent performance across a range of patient morphologies is required. In routine implementation, ULD protocols are typically defined using a reference patient model and subsequently adapted to patient size through adjustments in tube potential and corresponding dose levels. This context highlights that detectability-based optimization should be interpreted within a morphology-adaptive framework, rather than as a single global maximum criterion. The SSW- d' metric integrates detectability information across all investigated rods using Ω_i weighting coefficients that define their relative contributions. Depending on the clinical indication, the single-task SSW- d' formulation may be useful for focused assessments, such as the detection of herniated discs in lumbar spine CT examinations. However, most CT examinations involve multiple clinical tasks, and the SSW- d' metric was therefore extended to incorporate task-specific contrast weighting factors. Clinical indications such as lower back pain, suspected infection, or metastatic disease require adequate image quality across different anatomical structures and contrast conditions. In this context, unweighted aggregation of detectability values may be dominated by high-contrast rods, such as the bone-equivalent rod. This could potentially mask the contribution of low-contrast lesion detection tasks. The proposed weighting scheme mitigates this effect by rebalancing the relative contribution of each task. This preserves sensitivity to low-contrast detectability within the overall SSW- d' metric.

Consequently, the SSW- d' provides a task-based descriptor that more accurately reflects the various contrast requirements encountered in clinical CT examinations across different phantom diameters.

The resulting metric also enables comparison of imaging performance across acquisition and

reconstruction conditions. Task-based image quality assessments were performed on two CT systems for two 10-mm lesion detection tasks across a range of patient-equivalent diameters and tube voltages. For both CT systems, detectability decreased with increasing water-equivalent diameter, consistent with the expected impact of increased attenuation and scatter on image noise and contrast.³⁷ Differences in SSW-d' across acquisition conditions were observed between the two systems; however, these results should be interpreted within the context of the specific reconstruction settings, dose levels, and task definitions used in this study. In this framework, the photon-counting CT (PCCT) system showed higher SSW-d' values than the energy-integrating CT (EICT) system under the investigated conditions, with optimal performance observed at 120 kV for PCCT and Sn100 kV for EICT. These findings are consistent with previous reports describing improved task-based performance of PCCT in low-dose imaging applications.^{38,39,41} Importantly, SSW-d' reflects reconstructed image performance resulting from the combined influence of phantom diameter, acquisition parameters, reconstruction settings, detector technology, and task definition, and should not be interpreted as an intrinsic or generalizable ranking of scanner performance. Detectability was found to be strongly dependent on reconstruction kernels and algorithms, with variations of similar order to those observed between CT systems. This suggests that reconstruction strategy may influence task-based performance to a degree similar to hardware-related factors.

Overall, the SSW-d' metric highlights the potential for optimizing task-based image quality in ultra-low-dose CT protocols, where image degradation due to increased noise and reduced contrast is most pronounced, or alternatively for supporting further dose reduction strategies in such applications. These findings indicate that improvements in detectability are not solely driven by radiation dose but also strongly depend on reconstruction parameters, particularly the reconstruction kernel. In this context, SSW-d' provides a compact and clinically relevant framework for jointly assessing the effects of patient-equivalent diameter, imaging parameters, and reconstruction choices on lesion detectability in CT imaging.

This study has several limitations. First, only a limited number of two rod contrasts and phantom materials was investigated. Second, SSW-d' was not evaluated under clinically implemented tube current modulation (TCM). TCM implementations vary across manufacturers and may interact with patient size and non-linear reconstruction algorithms, potentially affecting noise and dose distributions, particularly in larger and heterogeneous phantoms. The use of fixed mA acquisitions and a simplified cylindrical phantom enabled controlled assessment of size-dependent effects, however,

these conditions do not fully reflect clinical acquisition strategies.⁴² Third, noise analysis relied on a 2D NPS approach, which does not account for out-of-plane correlations in 3D CT data; a full 3D NPS analysis could not be performed due to software limitations. Fourth, although the ULD acquisition and reconstruction parameters determined using SSW-d' showed promising performance in phantom experiments, their clinical generalizability across different patient morphologies remains to be established. While previous studies have demonstrated the clinical feasibility of ULD dose levels, the present work focuses on a proof-of-concept optimization framework based on SSW-d' under controlled conditions, which will require further validation in patient data. Fifth, while differences between CT systems were observed, these findings cannot be directly generalized to clinical performance. Further work is needed to evaluate SSW-d' across a broader range of CT technologies, reconstruction methods, and clinical tasks, in order to better understand the relative contributions of acquisition, detector, and reconstruction factors to overall image quality. Finally, an important next step will be to extend the current water-equivalent diameters-specific weighting approach by integrating dose-related weighting factors. This would allow SSW-d' to account simultaneously for variations in phantom diameter and radiation dose, potentially improving its applicability across a wider range of clinical acquisition protocols.

6 | CONCLUSION

A novel size-specific weighted detectability index (SSW-d') was developed to characterize reconstructed CT image performance while accounting for phantom-diameter variability and task contrast weighting. Applied to PCCT and EICT systems, the metric provided a unified task-based assessment of image performance across clinically relevant imaging conditions and proved useful for evaluating acquisition and reconstruction strategies in low- and ultra-low-dose CT.

6.1 | General formulation of the size-specific weighted detectability index (SSW-d')

The process of determining Size-Specific Weighted Detectability index (SSW-d') can be explained as follows: several rods with different contrast levels (C_i) in the phantom can be imaged at a single $CTDI_{vol}$ dose level and for different phantom water equivalent diameters ($D_i, i = 1, \dots, I$).

The d' values calculated for the different rods (d'_j with $j = 1, \dots, J$) and for different phantom water equivalent diameters were averaged according to the following

equation adapted from the method of Goppel et al⁶:

$$\langle SSW - d' \rangle_j = \frac{1}{2(D_l - D_1)} \sum_{i=1}^l K_i d'_{ij} \quad (11)$$

where $K_1 = D_2 - D_1, \dots, K_i = D_{i+1} - D_{i-1}$ ($i = 2, \dots, l-1$), $\dots, K_l = D_l - D_{l-1}$ are the diameter weights for the corresponding values of d'_{ij} .

The mean SSWD over all contrast rods was calculated as follows:

$$SSW - d' = \sum_{j=1}^J \langle SSW - d' \rangle_j \quad (12)$$

The weights Ω_j were given by the contrasts of the rods as follows:

$$\Omega_j = \frac{C_j^{-1}}{\sum_{j=1}^J C_j^{-1}} \quad (13)$$

This formulation was based on the approximately linear relationship between detectability and rod contrast at a fixed rod diameter. Equation (2) provides a reliable estimate of the weights, helping to strike a balance between the various inserts and preventing the easily detectable ones from dominating.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Authors will share data upon request to the corresponding author.

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SUPPORTING INFORMATION

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